AUTOMATED DAILY LITERATURE DIGEST • PhD RESEARCH WATCH

Particulate Matter
Monitoring & Health Effects

Issue 2026-10-10 | Entry window 10 October 2026 (full day) | 9 records screened in scope

9	4	0	1	3
RECORDS IN SCOPE (35 EXCLUDED)	OF 10 SUBTOPIC BANDS POPULATED	EFFECT ESTIMATES WITH A CI	TIER-A RECORDS	RECORDS CARRIED ON METADATA ALONE

Signal of the day

Sixty-six studies, ten years: low-cost PM sensors reach a median calibrated R^2 of 0.92 on the bench and 0.72 in the field, and median RMSE rises from 2.6 to 12.3 $\mu g\,m^{-3}$ on the same crossing.

A critical review of 66 peer-reviewed studies published 2015–2025 — 32 laboratory evaluations and 34 field deployments — puts numbers on the gap this watch keeps circling. Median calibrated R^2 was **0.92** ($n = 31$) in the laboratory against **0.72** ($n = 29$) in the field; median RMSE **2.6** $\mu g\,m^{-3}$ ($n = 7$) against **12.3** $\mu g\,m^{-3}$ ($n = 9$). The authors are careful that these summarise a heterogeneous literature rather than estimate a pooled effect: aerosol mixture, reference instrument, concentration range, averaging interval and validation design all differ between studies, and only 7 and 9 of the 66 reported RMSE at all. **Why it matters:** the review's conclusion is the operative one — calibration is not a correction applied once and carried unchanged into a new deployment. Machine-learning and hybrid models appear more often in field work and sometimes lower error, but independent validation across *time, place and sensor unit* is uneven, which is precisely the validation that a network, as opposed to a study, requires.

What changed today

- **The error figures rest on a thin subset.** Of 66 studies, **31** reported a calibrated R^2 and only **7** (laboratory) and **9** (field) reported RMSE. A median over nine field RMSEs is a weak estimate of what a network should expect, and the studies that report error at all are plausibly the ones that found it acceptable.
- **Microplastics travel with the resuspended mixture.** Two seasonal campaigns in central Los Angeles found airborne microplastics in every sample (2.2–4.0 MP m^{-3} cool season, 2.2–3.4 warm), fibre-dominated and concentrated in the 5–50 and 50–100 μm fractions, correlating at $\rho \geq 0.79$ with crustal, traffic and carbonaceous tracers while showing no consistent relationship with short-term meteorology. The inhalable fraction below 5 μm is not resolved.
- **A policy reform with emissions down and ambient flat.** Kazakhstan's 2021 Environmental Code cut reported stationary-source emissions by **5–6 %** between 2021 and 2024, yet no statistically significant improvement in ambient indicators appeared through 2025 and episodic exceedances persisted. With no control region, the null is as much a property of the design — and of a network too sparse to detect a 5 % change — as of the policy.
- **Trends that are mostly weather.** Eight years of Indian in-situ data give falling PM at several Indo-Gangetic sites (Sen's slope -0.893 to -0.097 $\mu g\,m^{-3}$ per month for $PM_{2.5}$), rising ozone (0.041 to 0.529), and rising Punjab. The paper's own finding — declining boundary-layer height and wind speed modulate all of it — is the reason those slopes are not emission trends.

Window continuity

The 9 October issue closed with `last_entry_date` at 9 October. This issue collects **10 October in full**, continuous with it; the 11 October issue follows on this run, and the **W41 weekly** (4–10 October) is filed under this date. Two PubMed-connector records dated 10 October by entry date also surfaced in the 11 October PubMed query and were rejected there as duplicates of this issue.

Corpus analytics

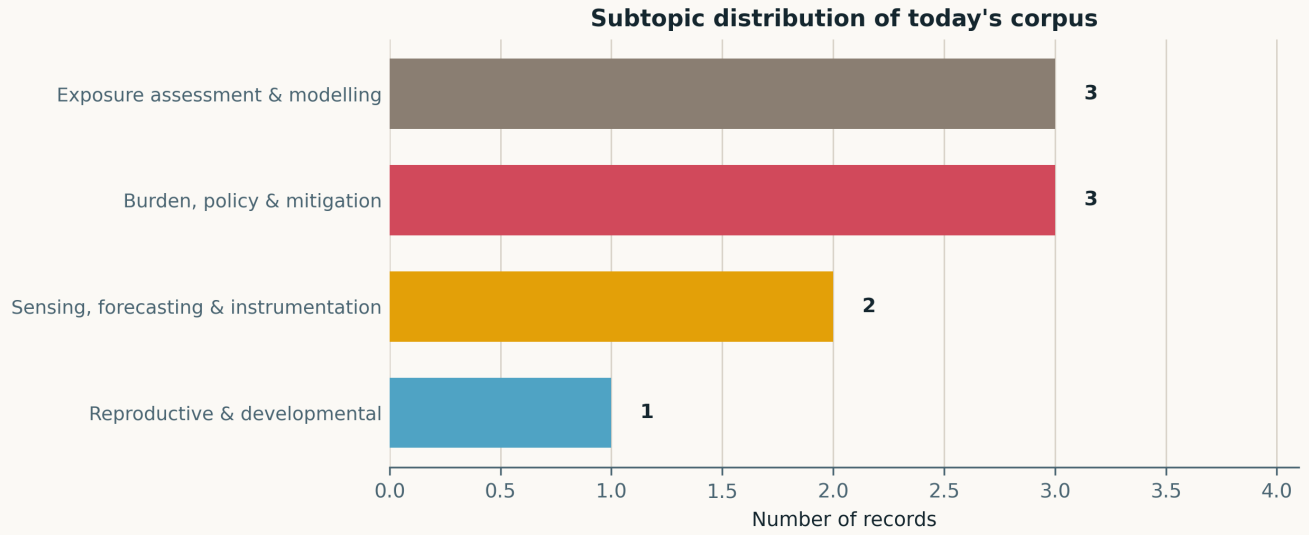


Fig. Subtopic assignment of the 9 in-scope records — a thin Saturday, the smallest issue since 4 October. **Four of ten bands populated**: *Exposure assessment* and *Burden* three each, *Sensing* two, *Reproductive* one. The six empty bands include every clinical band except reproductive, which is what a day with one health-endpoint record looks like.

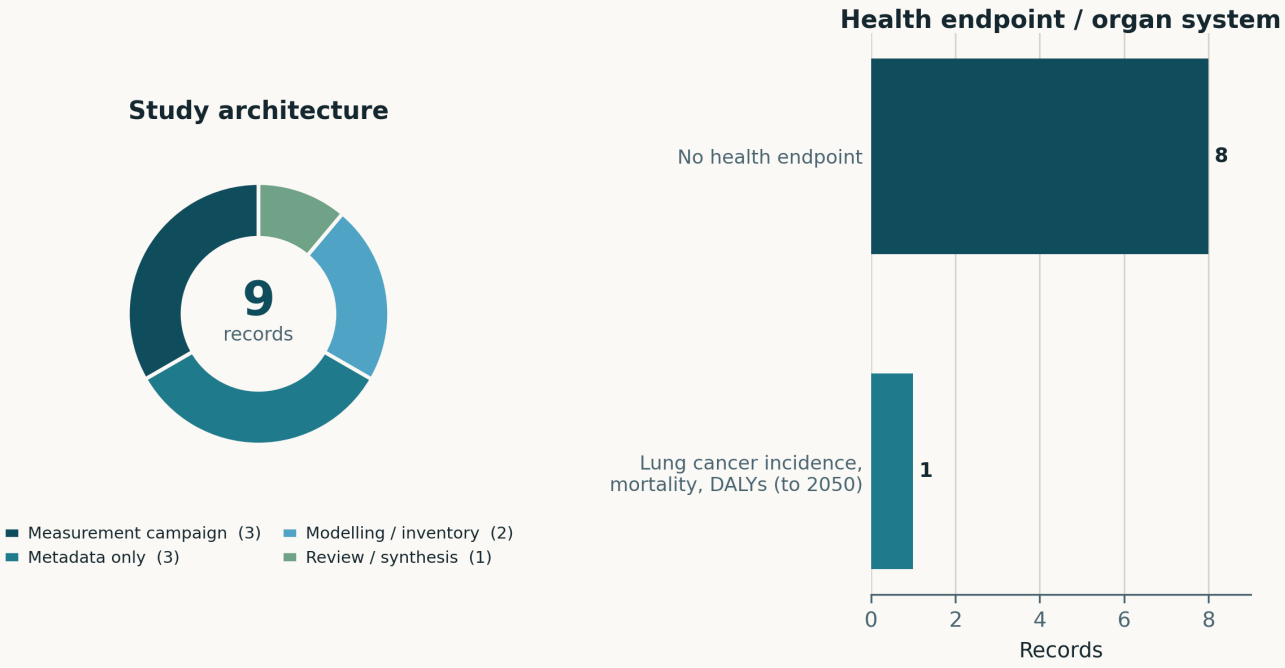


Fig. Left — architecture: three measurement campaigns, **three** metadata-only records (33 %), two modelling / inventory records and one review / synthesis (the sensor calibration review, the day's only tier-A record). Right — a single record reports a health endpoint (lung cancer burden, modelled rather than observed); one more asserts one by title only. Everything else reports concentrations, composition, model skill or measurement performance.

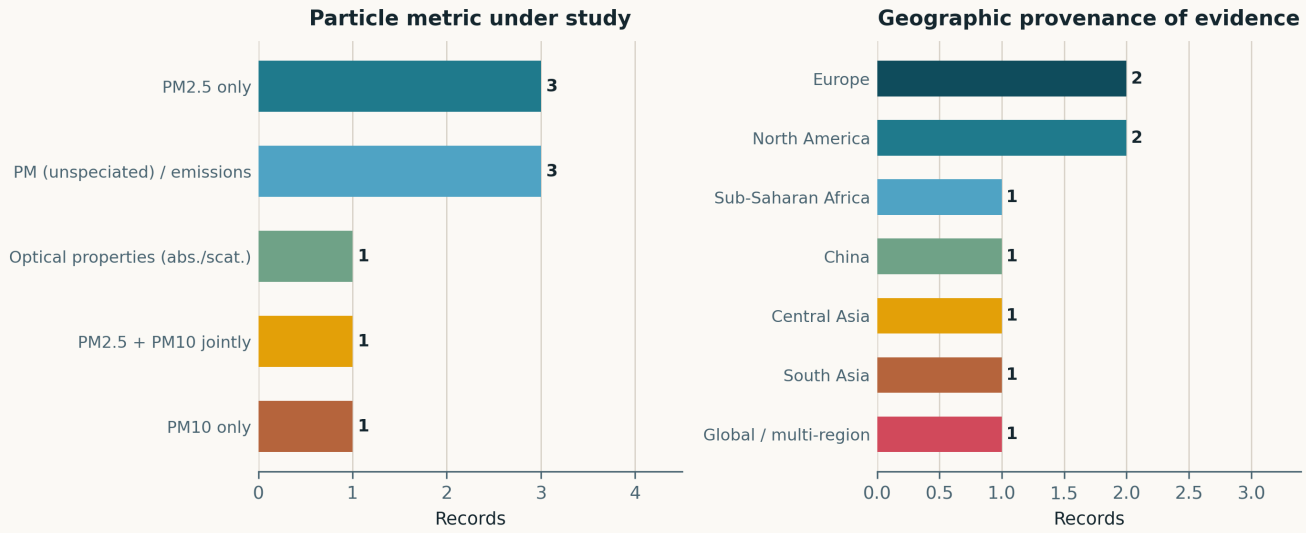


Fig. Left — particle metric: unspeciated PM or emissions **three**, $PM_{2.5}$ only three, and one each of PM_{10} only, $PM_{2.5}$ + PM_{10} jointly, and optical properties. Right — geography: North America **two** (the Los Angeles microplastics campaign and the China–US burden comparison), Europe two, and one each for China, South Asia, Central Asia, sub-Saharan Africa and global — the widest geographic spread per record of the backfill, on the fewest records.

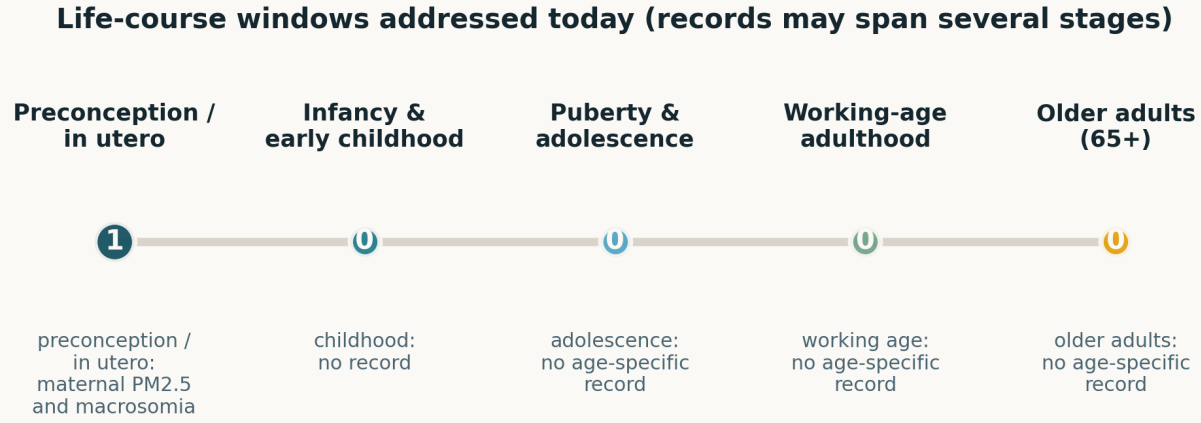


Fig. Life-course windows: one record only, in preconception / in utero (the nationwide maternal $PM_{2.5}$ and macrosomia study, carried on metadata alone). Childhood, adolescence, working age and older adults have no age-specific record this issue — the lung-cancer burden projection spans all ages and is not assigned to a window.

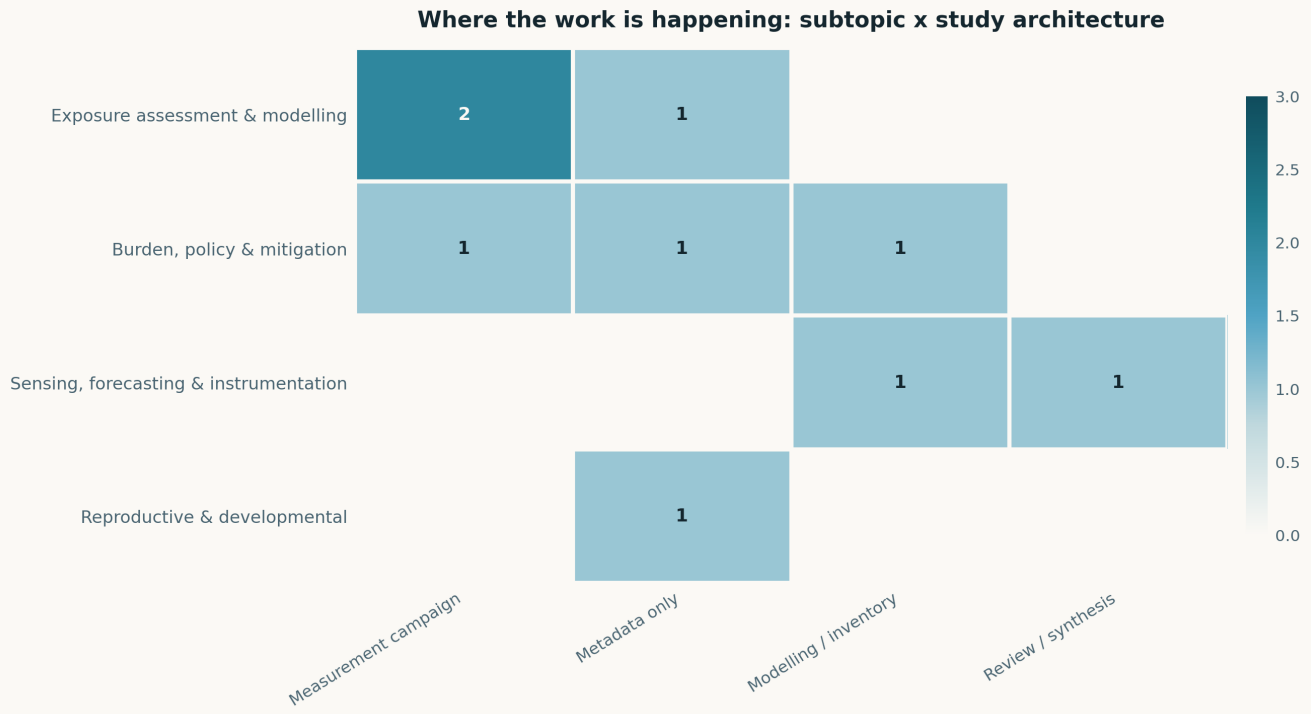


Fig. Subtopic × study architecture for the 9 records. With nine records across four subtopics and five architectures, no cell holds more than two; the matrix is nearly diagonal, which is what a thin and heterogeneous day looks like rather than any structure in the field.

No forest plot this issue. No admitted record reports a PM effect estimate with a confidence interval. The calibration review's headline quantities are measurement-performance statistics (median R^2 and RMSE across studies), not effect estimates, and pooling them would misrepresent a literature its own authors decline to pool; the lung cancer projection reports burden rather than an exposure-response estimate. The f5 panel is absent by design, not by build failure.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Records marked *metadata only* carry no recoverable abstract and assert no finding.

Reproductive & developmental — 1 record

Qiang maternal PM macrosomia 2026 Chin Med J (Engl)

Interaction of maternal $PM_{2.5}$ exposure and psychosocial state on newborn macrosomia: A nationwide population-based study

METADATA ONLY. Metadata-only record: the publisher deposited title, authors and DOI on 10 Oct with no abstract, and the retry chain recovered none. By title, a nationwide study of the interaction between maternal $PM_{2.5}$ exposure and maternal psychosocial state on newborn macrosomia. Interaction claims between a modelled environmental exposure and a self-reported psychosocial measure are the design most prone to residual confounding by socioeconomic position, and macrosomia is a thresholded birthweight outcome whose prevalence is sensitive to that. Relevance: moderate - birth cohorts are where personal rather than residential exposure measurement would change the estimate most. On the abstract-retry list.

doi: 10.1097/cm9.0000000000004256

Exposure assessment & modelling — 3 records

Fazelvalipour LA microplastics 2026 Preprints.org (preprint)

Size-Resolved Characteristics and Source Influences of Airborne Microplastics in Central Los Angeles

Two seasonal campaigns in central Los Angeles: microplastics were in every sample at 2.2-4.0 MP/m3 (cool season) and 2.2-3.4 MP/m3 (warm), dominated by fibres and by the 5-50 and 50-100 um size fractions. Total MP correlated strongly (rho >= 0.79) with crustal (Fe, Si, Mn, Ca, Al, Mg), traffic (Ba, Cu, Ni) and carbonaceous (OC, EC) tracers, while short-term meteorology showed no consistent relationship, implying MPs travel as part of the resuspended urban particle mixture rather than as a separate contaminant. Limitation: a preprint; microscopy-based MP counts carry large operator and blank uncertainty, correlation with crustal tracers is also what resuspension artefact looks like, and particles under 5 um - the inhalable fraction - are not resolved. Relevance: moderate-high - supports the same-day tyre-wear argument that the growing non-exhaust, coarse, resuspended fraction is what networks increasingly measure badly.

doi: 10.20944/preprints202610.0577.v1

Chaudhary India meteorology 2026 Environ Monit Assess

Meteorological modulation of ground-level air quality in India based on in situ measurements

Eight years of in-situ measurements across Indian monitoring sites with Sen's slope trends. $PM_{2.5}$ fell at several Indo-Gangetic sites (-0.893 to -0.097 $\mu g\,m^{-3}$/month) and PM_{10} likewise (-1.320 to -0.169), while Punjab rose on local sources; ozone increased (0.041 to 0.529 $\mu g\,m^{-3}$/month), read as aerosol inhibition weakening plus more precursors. Declining boundary-layer height and wind speed over north and east India modulate all of it. Limitation: Sen's slope on eight years of CPCB data cannot separate emission change from meteorological trend, which is the paper's own point turned against its trend estimates; station siting and instrument changes over that period are undocumented. Relevance: moderate - the regional version of the same-day argument that a concentration trend is not an emissions trend.

doi: 10.1007/s10661-026-15915-9

Anwar Pretoria FTIR 2026 Aerosol Air Qual Res

$PM_{2.5}$ Composition and Organic Matter Sources in Pretoria, South Africa Employing Fourier Transform-Infrared Spectroscopy

METADATA ONLY. Metadata-only record: title, authors and DOI deposited 10 Oct with no abstract. By title, FT-IR functional-group analysis of $PM_{2.5}$ organic matter in Pretoria with source attribution. FT-IR on routine filters is one of the few speciation routes cheap enough to run network-wide, and sub-Saharan African composition data are scarce enough that any new site matters. Relevance: high for the composition gap this corpus repeatedly hits; on the abstract-retry list.

doi: 10.1007/s44408-026-00172-9

Sensing, forecasting & instrumentation — 2 records

Fang LCS calibration review 2026 Curr Environ Health Rep

Calibration of Low-Cost Particulate Matter Sensors for Exposure Assessment: A Critical Review

Sixty-six studies, 32 laboratory evaluations and 34 field studies. Median calibrated R2 was 0.92 (n = 31) in the laboratory against 0.72 (n = 29) in the field, and median RMSE 2.6 $\mu g\,m^{-3}$ (n = 7) against 12.3 $\mu g\,m^{-3}$ (n = 9) - a factor of nearly five in error on crossing from bench to deployment. The authors are explicit that these are summaries of a heterogeneous literature, not a pooled effect: aerosol mixture, comparison instrument, concentration range, averaging interval and validation design all differ. RH and temperature were handled in most studies; ML and hybrid models appear more often in field work and sometimes lower error, but independent validation across time, place and sensor unit is uneven. Limitation: only 7 and 9 studies reported RMSE at all, so the headline error figures rest on a small and probably favourable subset. Relevance: high - this is the reference statement that calibration does not transfer, which is the standing assumption behind every distributed network claim in this corpus.

doi: 10.1007/s40572-026-00566-2

Chiappini WRF-Chem nudging 2026 Atmosphere

Regime-Dependent Effectiveness of Spectral Nudging in WRF-Chem: Meteorological and Saharan Dust Skill over the Central Mediterranean

A two-tier design separates the meteorological response to spectral nudging (a 104-day chemistry-free pair) from the dust-transport response (paired event simulations). Nudging improves sea-level pressure most (daily RMSE 2.08 to 1.53 hPa) and 10 m wind less (2.50 to 2.20 m/s), while 2 m temperature does not improve (2.05 to 2.15 C); the benefit recurs episodically rather than growing with simulation age. Dust skill is conditional: in the weakly forced June case all seven AERONET sites improved (station-mean RMSE 0.457 to 0.235, correlation 0.778 to 0.881), while in March RMSE fell at all eight sites but mean correlation was unchanged. Limitation: two outbreaks is a thin basis for a regime claim, and AOD is a column integral that constrains surface dust PM only weakly. Relevance: moderate - dust is the event class where satellite-and-model PM products diverge most from surface nodes.

doi: 10.3390/atmos17100992

Burden, policy & mitigation 3 records

Mukhamediyarov Kazakhstan code 2026 Atmosphere

Air Quality and Emission Trends Following the 2021 Environmental Code Reform in Kazakhstan Across Regions and Industrial Sectors

A before-and-after comparison of Kazakhstan's 2021 Environmental Code reform, which tightened standards and pushed Best Available Techniques. Reported stationary-source emissions fell roughly 5-6% between 2021 and 2024, but no statistically significant improvement in assessed ambient indicators appeared between 2021 and 2025, and episodic PM exceedances persisted in several cities. Limitation: no control region and no counterfactual, so the null is as much about the design as about the policy; reported emissions are self-declared inventory, and the ambient network is sparse enough that a 5% emission change would be undetectable even if real. Relevance: moderate - a clean example of why policy evaluation needs measurement density the regulatory network does not have; the 7 October congestion-pricing record is the version that had it.

doi: 10.3390/atmos17100993

Guo lung cancer burden 2026 Chin Med J Pulm Crit Care Med

Trends in lung cancer burden, risk factors, and disease projections in China and the United States, 1990-2023

GBD 2023 data with joinpoint regression, three-factor Das-Gupta decomposition and Bayesian age-period-cohort projection to 2050, comparing lung cancer burden and its attributable risks - smoking, air pollution, occupational exposure, metabolic and dietary factors - between China and the United States. Limitation: GBD attribution for air pollution is a modelled counterfactual built on exposure surfaces and borrowed concentration-response functions, so the pollution share is an assumption propagated forward, not an observation; the abstract reports no attributable fractions. Relevance: low-moderate - useful as the burden framing that the measurement side of this corpus feeds, but it adds no exposure information of its own.

doi: 10.1016/j.pccm.2026.08.006

Abdella Italy urban-rural 2026 Atmos Environ

Cleaning the air or shifting the burden? Characterizing urban-rural disparities in air pollution-attributable mortality across Italy

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 10 Oct with no abstract. By title, an attributable-mortality analysis asking whether Italian air-quality gains have been distributed or merely displaced from urban to rural populations. Rural exposure is systematically the worse-measured half of any national network, so a rural-urban disparity estimate inherits the monitoring asymmetry it is trying to describe - the 8 October US block-level record showed spatial aggregation attenuating exactly this kind of disparity. Relevance: moderate-high; on the abstract-retry list.

doi: 10.1016/j.atmosenv.2026.122420

Corpus register

First author	Focus	Journal	Design	Metric	Region
Reproductive & developmental					
Qiang maternal PM macrosomia	Interaction of maternal $PM_{2.5}$ exposure and psychosocial state on newborn macrosomia: A nationwide population-based study	*Chin Med J (Engl)*	Metadata only (no abstract)	Metadata only - maternal $PM_{2.5}$ exposure with psychosocial state, nationwide (by title)	China (nationwide)
Sensing, forecasting & instrumentation					
Chiappini WRF-Chem nudging	Regime-Dependent Effectiveness of Spectral Nudging in WRF-Chem: Meteorological and Saharan Dust Skill over the Central Mediterranean	*Atmosphere*	Chemical transport model	Modelled Saharan dust AOD550 against AERONET; meteorological RMSE from nudged vs free runs	Europe (Central Mediterranean)
Fang LCS calibration review	Calibration of Low-Cost Particulate Matter Sensors for Exposure Assessment: A Critical Review	*Curr Environ Health Rep*	Systematic review (PRISMA)	Calibrated R2 and RMSE of low-cost PM sensors against reference instruments, laboratory vs field	Global (66 studies, 2015-2025)
Exposure assessment & modelling					
Anwar Pretoria FTIR	$PM_{2.5}$ Composition and Organic Matter Sources in Pretoria, South Africa Employing Fourier Transform-Infrared Spectroscopy	*Aerosol Air Qual Res*	Metadata only (no abstract)	Metadata only - FT-IR functional-group speciation of $PM_{2.5}$ organic matter (by title)	South Africa (Pretoria)
Chaudhary India meteorology	Meteorological modulation of ground-level air quality in India based on in situ measurements	*Environ Monit Assess*	Monitoring-record analysis	PM_{10}, $PM_{2.5}$, O3, NO2, SO2 with PBL height, RH, temperature, precipitation, wind, surface pressure (2017-2024)	India (national, Indo-Gangetic Plain focus)
Fazelvalipour LA microplastics	Size-Resolved Characteristics and Source Influences of Airborne Microplastics in Central Los Angeles	*Preprints.org (preprint)*	Field measurement	PM_{10}-associated airborne microplastics (counts/m3, morphology, size) with PM_{10} composition	United States (central Los Angeles)
Burden, policy & mitigation					
Abdella Italy urban-rural	Cleaning the air or shifting the burden? Characterizing urban-rural disparities in air pollution-attributable mortality across Italy	*Atmos Environ*	Metadata only (no abstract)	Metadata only - urban-rural disparity in PM-attributable mortality across Italy (by title)	Italy (national)
Guo lung cancer burden	Trends in lung cancer burden, risk factors, and disease projections in China and the United States, 1990-2023	*Chin Med J Pulm Crit Care Med*	Health impact model	GBD 2023 risk-factor attribution including ambient and household air pollution	China and United States
Mukhamediyarov Kazakhstan code	Air Quality and Emission Trends Following the 2021 Environmental Code Reform in Kazakhstan Across Regions and Industrial Sectors	*Atmosphere*	Monitoring-record analysis	Stationary-source emissions ($PM_{2.5}$, SO2, CO, NO2) and ambient indicators by region and sector	Kazakhstan (national)

Provenance and method

Entry window **10 October 2026**, single day, continuous with the 9 October issue. Harvester legs: **PubMed** health **1**, sensing **0** — the thinnest PubMed day of the series, a Saturday entry date; **Europe PMC 20** (8 after the retro-index gate dropped 12); **Crossref by ISSN 32**; **arXiv** returned no entry dated 10 October; **OpenAlex** HTTP 429. 41 raw records, no cross-source duplicates, **41** fresh. The **PubMed connector** (`[EDAT]` 9–11 October) returned 9 PMIDs dated 10 October, 5 already carried and **3** appended — **44** candidates screened, **9 admitted**, **35 rejected** with logged reasons. **Three records carried on metadata alone** (33 %); the abstract-retry chain recovered **3 of 7** attempted, all three via PubMed, and none of the Elsevier or Springer deposits. **Consensus** returned 10 calibration hits, **0 new** — all pre-date the window and several are cited by the calibration review admitted here. **Scholar Gateway** fails with `ACCESS_DENIED`. **ClinicalTrials.gov**: one new registration in the 4–11 October window (EPIC-AIR, NCT07500948), reported in the W41 weekly. Abstracts remain the copyright of their respective publishers.